



Challenges of the assessment of nature conservation value of aquatic habitats using microalgae

Judit Görgényi¹ · Enikő T-Krasznai¹ · Zsuzsanna Kókai¹ ·
Viktória B-Béres¹ · Gábor Várbíró¹ · Éva Ács¹ · Keve Tihamér Kiss ·
Gábor Borics · Áron Lukács²

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Abstract The increasing focus on ecosystem restoration in European environmental policy highlights the need for reliable methods to quantify the nature conservation value (NCV) of aquatic habitats. Current Water Framework Directive (WFD) metrics assess ecological quality mainly through the phytoplankton community composition and abundance.

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J. Görgényi (✉) · E. T-Krasznai · Z. Kókai · V. B-Béres ·
G. Várbíró · G. Borics · Á. Lukács
Institute of Aquatic Ecology, Functional Algology
Research Group, HUN-REN Centre for Ecological
Research, Bem Square 18/C, Debrecen 4026, Hungary
e-mail: gorgenyi.judit@ecolres.hu

É. Ács · K. T. Kiss
Department of Aquatic Environmental Sciences, Faculty
of Water Sciences, Ludovika University of Public Service,
Ludovika Square 2., Budapest 1083, Hungary

É. Ács · K. T. Kiss
National Laboratory for Water Science and Water Security,
Budapest, Hungary

Á. Lukács
University of Helsinki, Lammi Biological Station,
Pääjärventie 320, 16900 Lammi, Finland

However, these approaches do not reflect properly the real conservation value of the habitats, because they overlook the contribution of rare and unique algal species. Currently, the method proposed by Coesel is the only microalgae-based framework explicitly incorporating rarity, using desmid assemblages to assess the conservation value of aquatic habitats. In this study, we investigated its applicability to the Hungarian aquatic ecosystems using a database containing 379 desmid species from 101 localities, representing 14 habitat types. We calculated the NCV value of all localities. The highest NCV values were found in old bog lakes, marshlands, oxbows and large saline lakes, while reservoirs, large rivers and heavily modified habitats showed lower scores. No significant relationship was detected between the NCV value and the WFD-compliant Ecological Quality Ratio (EQR) metrics, indicating that ecological status metrics are weak predictors of conservation value. Although the desmid-based NCV method provides a promising foundation, its components require refinement and habitat-specific calibration to support more robust conservation assessments.

Keywords Microalgae · Nature conservation value · Desmids

Introduction

Freshwater ecosystems are among the most valuable yet vulnerable habitats on Earth (Dudgeon & Strayer, 2025). Although they cover only about the 1% of the Earth's surface, they support nearly 10% of all known animal species (Balian et al., 2008; Strayer & Dudgeon, 2010). In addition, freshwater systems provide a wide range of ecosystem services essential to human well-being, including clean water, food resources and recreational opportunities. At the same time, freshwaters are under multiple pressures due to pollution, overexploitation, habitat degradation, biological invasion and climate change; as a result, freshwater biodiversity is declining rapidly (Tockner, 2021). In response to these escalating threats, the European Union introduced the Water Framework Directive (WFD) in 2000, establishing an integrated approach to water protection with the objective of achieving at least 'good' ecological and chemical status for all surface and groundwater bodies by 2015 (WFD: 2000/60/EC). Over the past 25 years, the WFD has become a cornerstone of European water policy, addressing key challenges such as pollution, habitat degradation and the impacts of climate change (Adina et al., 2024). The WFD takes an environmental approach, assessing water bodies on the basis of physical and chemical parameters and selected five biological quality elements: phytoplankton, phyto-benthos, macrophytes, macroinvertebrates and fish. Assessment methods vary substantially across types (lakes, rivers, reservoirs) and elements: phyto-benthos assessment predominantly uses diatom-based weighted average metrics like the Indice de Polluosensibilité Spécifique (Coste, 1982); macrophytes employ indices such as the Integrated Macrophytes Qualification Index incorporating species composition and coverage (Szilágyi et al., 2018); macroinvertebrates utilize composition/abundance, richness/diversity and sensitivity/tolerance indices (Hering et al., 2006); fish assessment applies the Index of Biological Integrity, while phytoplankton relies on species composition and chlorophyll-a concentration (Padisák et al., 2006; Carvalho et al., 2013). Classification necessarily varies by water body type (rivers, lakes, transitional, coastal waters) and type-descriptor variables (alkalinity, humic content, depth, size,

altitude, etc.). These types reflect legitimate ecological variation rather than methodological inadequacy (Solheim et al., 2008).

Three main types of pressures are generally identified in water bodies: organic and inorganic nutrient loads (e.g. fertilizers, sewage), moreover the hydromorphological loads (physical changes to a water body). Organic and inorganic nutrient loads can lead to eutrophication (Kováčová, 2022), while hydromorphological modifications change the structures of water body, affecting the habitat complexity and flow patterns (Elosegi et al., 2010).

Autotrophic primary producers, (plants and microalgae) are responsible for the organic matter production in waters (Wetzel, 2001), while animals and heterotrophic microbes are essential for decomposing organic compounds back into inorganic forms (Kirchman, 2018). Consequently, the WFD proposes to use primary producers (macrophytes, phytoplankton, phytobenthos) as indicators of inorganic nutrient enrichment, while macroinvertebrates and fish to indicate both organic nutrient pressures and hydromorphological alterations (Solheim et al., 2013). We note here that several divisions of microalgae contain mixotrophic organisms that consume organic matter in dissolved (euglenophytes) or even in particulate (chrysophytes, dinoflagellates) forms and they respond rapidly to the conditions that characterize waters in different levels of organic pollutions. These algae were involved in many saprobic systems (Pantle & Buck, 1955; Zelinka & Marvan, 1961) and used as effective bioindicators of water quality (Ugya et al., 2025). Building on these principles, the member states of the EU developed stressor-specific metrics that provide a robust and reliable framework for characterizing the ecological status of water bodies, as they accurately reflect the different types and intensity of environmental pressures (Poikane et al., 2014). The ecological quality ratios (EQRs) obtained by these indices not only enable the diagnosis of pressures but also provide a basis for the planning and implementing the necessary environmental interventions and measures by which the good ecological status of water bodies can be maintained or achieved.

After a long-lasting debate, a new legal act, the Nature Restoration Law (NRL) was finally adopted

in the European Union. The law (Regulation, 2024/1991) adopted by the European Parliament and the Council of the European Union as part of the European Green Deal, represents the first legally binding, union-wide framework explicitly aimed at the restoration of degraded ecosystems. The regulation entered into force in 2024, and the aim is to reverse decades of biodiversity loss, habitat degradation and ecosystem fragmentation, which presently affect approximately 80% of habitats across the European Union. This newly introduced act focuses not only on the terrestrial (Agricultural, Peatland, Forest or Urban) ecosystems but also on the aquatic ecosystems (Marine and Coastal Ecosystems, River and Freshwater Ecosystems) with special attention to the presence or absence of the rare, threatened, native species. For aquatic ecosystems, member states are required to implement restoration measures on at least 30% of degraded areas by 2030, increasing this ratio to 60% by 2040 and 90% by 2050, with the overarching aim of bringing at least 90% of habitat areas into good ecological condition and maintaining the favourable reference areas for each habitat type. It must be ensured that habitats once restored do not deteriorate further (non-deterioration requirement). These measures collectively contribute to biodiversity conservation, climate mitigation through enhanced carbon sequestration and storage, improved water and soil regulation and the reduction of natural hazards such as floods and droughts. Within river and wetland systems, the regulation mandates the restoration of 25000 km of watercourses to free-flowing conditions by 2030. Member states must prioritize the removal of obsolete hydromorphological barriers, compile inventories of artificial barriers targeted for removal and implement actions to restore and maintain natural floodplain dynamics and longitudinal connectivity. In addition, drained peatlands must be restored through partial or full rewetting where appropriate by 2030 (Regulation, 2024/1991).

Both the Water Framework Directive and the Nature Restoration Law involve highly complex tasks and many significant ecological inventions and interventions. Although they show significant overlap in their intention to restore freshwater ecosystems (such as preserving the natural ecological state, mitigating degradation or ensuring long-term ecosystem sustainability), there are fundamental conceptual differences in

their specific objectives and management approaches. While the WFD focuses on overall ecological status of the water bodies, the NCL prioritizes specific species and habitat protection (Ciscato et al., 2025). Despite these differences, both frameworks face significant challenges in developing robust, comparable ecological metrics by which efficiency of restoration interventions can be quantified and assessed.

The key question of our study is as follows: how can the composition of microalgal communities be utilized to quantify the nature conservation value of aquatic habitats? To date, only one nature conservation index based on microalgae as indicator organisms has been developed, namely the Nature Conservation Value (NCV) proposed by Coesel over twenty years ago (Coesel, 1998). This index has been based exclusively on desmid assemblages, and it was originally designed for small, natural standing waters such as marshes and bogs. Based on a previous floristic survey (Fehér, 2007), Krasznai and her colleagues adapted the Coesel's original method to Hungarian conditions and assessed the conservation value of the oxbow lake (2008). Apart from that, however, there were no further attempts to apply the method, particularly on a larger spatial scale. Reflecting the achievements of the NRL, this study aims to evaluate the applicability of Coesel's NCV framework to Hungarian aquatic systems.

- We hypothesized that NCVs show considerable differences among the habitat types, reflecting the ecological heterogeneity of the Hungarian aquatic ecosystems.
- We also hypothesized that WFD-compliant methods provide a good estimate of NCVs.

Materials and methods

Data sources

The microalga database for Hungary was compiled from multiple sources. Primary data were extracted from algal handbooks that included both taxonomic descriptions and occurrence data (Felföldy, 1972, 1981, 1985; Uherkovich et al., 1995; Németh, 1997; Schmidt & Fehér, 1998, 1999, 2001; Grigorszky et al., 1999). The database was complemented with data from recent floristic studies published in the scientific literature, as well as from grey literature

(unpublished doctoral theses, governmental reports). Taxonomic names were carefully checked and standardized using AlgaeBase, and where necessary, updated to currently accepted nomenclature (Guiry, 2010). Synonyms having similar names after the updates of the names were merged.

In addition, phytoplankton monitoring data provided by the Hungarian General Directorate of Water Management were also incorporated in the database. The resulting species-by-site matrix consists exclusively of planktic and metaphytic taxa; benthic diatoms were not included.

Habitat types

Habitat types were delineated using a set of type-descriptor variables. These included water category (surface or subsurface waters; flowing or standing waters), chemical characteristics (organic, high alkalinity, moderate alkalinity), origin (natural or artificial), hydrological regime (permanent or temporary), size ($> 10 \text{ km}^2$) and depth (deep habitats defined as having a mean depth $> 4 \text{ m}$). Based on these criteria, a total of 13 habitat types were distinguished. The complete habitat typology is presented in ESM 1.

Nature conservation value (NCV)

The Nature Conservation Value (NCV) has been proposed by Coesel (1998) to assess the ecological status of small, natural standing waters (bog lakes and marshlands). This method is based on the composition of desmid communities, as these algae are highly sensitive to changes in water chemistry, trophic status and habitat integrity that makes them reliable bioindicators of clean, oligotrophic freshwater ecosystems (Celewicz et al., 2022).

The NCV is calculated as the sum of three components: species diversity (D), rarity of taxa (R) and ecosystem maturity (M), each reflecting a different ecological characteristic of desmid assemblages. As a measure of diversity (D), Coesel proposed to apply the species richness, i.e. the total number of species in the sample. The measure of ecosystem maturity (M) reflects the degree to which a species is associated with mature and ecologically stable ecosystems. Each species has been evaluated on a four-point scale.

Species assigned a value of 1 are characterized by a broad ecological tolerance and are commonly found in pioneer communities or disturbed habitats. In contrast, species with a value of 4 are considered highly indicative of well-structured, balanced ecosystems and are typically associated with advanced stages of ecological succession. Higher values therefore indicate a stronger relationship with ecosystem maturity and ecological integrity. In terms of rarity (which reflects how rare a species is in a region or country), species were evaluated on a three-point scale ranging from 1 (common) to 3 (very rare) (ones). Finally, all three values (D , cumulative values of M and R) were categorized into four quality classes (D : 1–3; R : 1–3; M : 1–4) applying habitat-type-specific boundaries. Summing of the three measures means that the theoretical minimum of NCV is one, while its maximum value is ten.

Higher NCV values indicate water bodies hosting numerous rare and sensitive desmid species, corresponding to clean, stable and undisturbed conditions with high conservation value. In contrast, lower NCV values generally typically reflect degraded, disturbed and nutrient-rich habitats dominated by common, tolerant species and therefore indicate low conservation value. However, it should be noted that low NCV values may occasionally occur in species-poor, Sphagnum-dominated acidic and oligotrophic peat bogs, which nevertheless represent ecologically valuable habitats (Neustupa et al., 2023).

The objective of our study was to evaluate the applicability of the Coesel method to Hungarian aquatic ecosystems and to determine which localities and habitat types exhibit the highest conservation value when assessed through this framework. Therefore, we applied the Coesel method for assessing the nature conservation value at both the locality and habitat-type levels. As calcareous bedrock predominates in Hungary, all water bodies fall within Coesel's neutral to alkaline type ($\text{pH} > 6.5$) (Coesel, 1998). We note here that the vast majority of Hungarian surface waters are alkaline and characterized by Ca-HCO_3 -dominated water chemistry. In peat lakes that develop within these naturally alkaline systems, peat formation and biological activity may occasionally result in somewhat lower pH values in the immediate vicinity of the vegetation. However, even in these habitats, the pH generally remains between 6 and 7 (Borics et al., 2003). In our dataset, pH values were mostly

around 6.5. Therefore, we considered that applying a threshold of pH 6.5 would artificially split a relatively homogeneous group of habitats. For this reason, we did not apply indicator values that were developed specifically for low-pH habitats.

Accordingly, we applied the boundaries defined for these water types. A comprehensive table of all identified taxa was compiled, including the Rarity (r) and Maturity (m) values for each taxon. These standardized values were used in the calculation of NCV value (ESM 2). It should be noted that taxa identified at the “varietas” level in the national database were excluded from the NCV analysis because Coesel’s rarity and maturity values are available only at the species level.

Differences among water body types were indicated using boxplots, while differences among localities were illustrated using grouped bar plots, in which the localities were arranged according to habitat type. Both boxplots and bar plots were generated using R statistical programming (R Core Team, 2025).

Comparison of NCV and EQR values

The phytoplankton-based, WFD-compliant Hungarian method assesses water bodies on a scale from 0 to 1, where 1 indicates high ecological status (near-reference conditions), and values closer to 0 indicate degraded conditions (Borics et al., 2018). Benthic diatom-based assessment was done on basis of Kelly et al. (2014).

We calculated the Nature Conservation Value (NCV) as well as the phytobenthos- and phytoplankton-based Ecological Quality Ratio (EQR) values for Hungarian water bodies larger than 50 hectares. In total, 50 localities were included in the study, representing eight habitat types (rivers, canals, oxbows, reservoirs, soda pans, pit lakes, large saline lakes and lakes).

The relationship between NCV and EQR values was evaluated using scatter plots with fitted linear regression lines. The strength of the relationship was quantified using the correlation coefficient (R). Statistical significance was considered when $P < 0.05$.

Results

The database

The database contains altogether 379 desmid species, occurred in 101 localities (ESM 3). The most frequent taxon was the *Closterium acerosum* Ehrenberg ex Ralfs (Charophyta, Zygnematophyceae), which occurred in 38 localities. The number of species that could be found exclusively in a single locality was 138, which comprises 36.8% of the Hungarian desmid flora. Species richness varied markedly among sampling sites. The highest number of species was observed in Lake Baláta (118 species), whereas there were locations where no species were found. Among habitat types, bog lakes (252 species) and oxbows (129 species) contain the highest number of desmid species, while reservoirs (8 species) and springs (6 species) had the lowest numbers (Table 1).

Differences in the NCV values among habitat types

Although each habitat type included several localities, NCV values could not be calculated for all of them, because desmid species were absent from many sites. Therefore, only evaluable localities (i.e. those containing desmids) were included in the analysis. The NCV values varied considerably both across (Fig. 1) and within (ESM 4) the habitat types. The highest median NCV values (NCV = 7) were recorded in large saline lakes, marshlands and old bog lakes, followed by pit lakes, oxbows and natural lakes (NCV = 6). Soda pans, ponds and canals showed moderate conservation value (NCV = 5), whereas streams, large rivers, reservoirs and fishponds exhibited lower values (NCV = 4). Springs had the lowest NCV value (NCV = 3) (Fig. 1).

The highest variation could be observed in the case of oxbows (NCV=3–9) and bog lakes (NCV=4–10). The values showed smaller but still large changes in the case of rivers (NCV=2–6). Considerably smaller variation in NCV values was characteristic for the other types where the maximum NCV values ranged between 6 and 8 (ESM 4).

Table 1 Number of desmid species found in the different habitat types

Habitat types	Number of desmid species
Springs	6
Streams	22
Rivers	106
Canals	23
Oxbows	129
Reservoirs	8
Soda pans	57
Bog lakes	252
Ponds (<1 ha)	54
Fishponds	41
Marshlands	42
Pit lakes	20
Large saline lakes	53
Lake	78

Relationship between ecological quality (EQR) and nature conservation values (NCVs) at the level of localities

At locality level, no relationship was detected between NCV and EQR values (Fig. 2a, b). The coefficient of determination was extremely low for both phytoplankton-based ($R = 0.035$) and phytoenthos-based EQRs ($R = 0.16$). However, two interesting patterns emerged. The scatter of data indicated that higher NCV values (> 6) could not be observed when the phytoplankton EQRs were lower than 0.5. Similarly, in the case of the phytoenthos at high EQR values (> 0.76), the NCVs did not reach lower values than 4.

When this relationship was examined within habitat types, only two cases (rivers and oxbows) had sufficient data for evaluation (i.e. more than three localities with available EQR values within the habitat type) (Fig. 3a, b). In rivers and oxbow lakes,

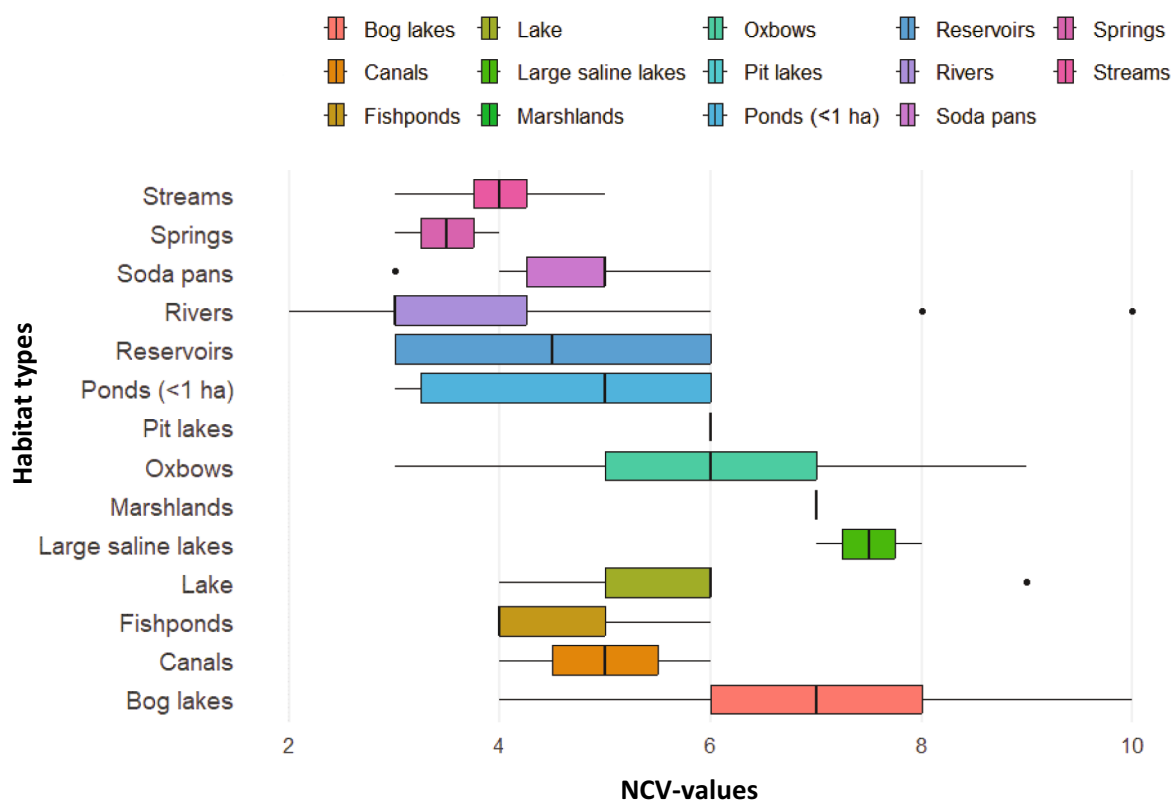


Fig. 1 Distribution of NCV values in the different habitat types. The centre line within the box represents the median; the upper and lower edges of the box represent the lower and

upper quartiles and circles are the outliers. The coloured boxes at the top indicate the different habitat categories

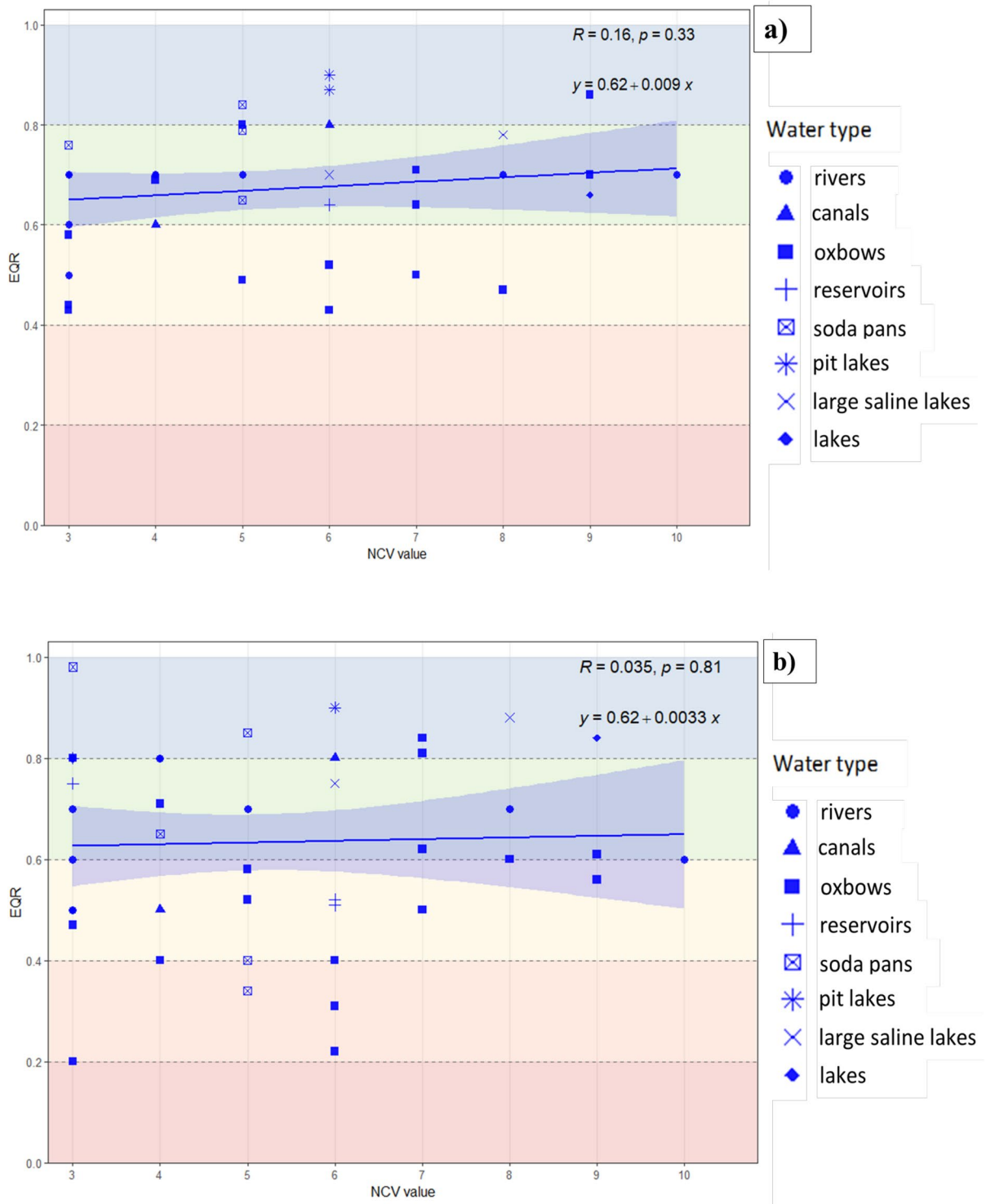
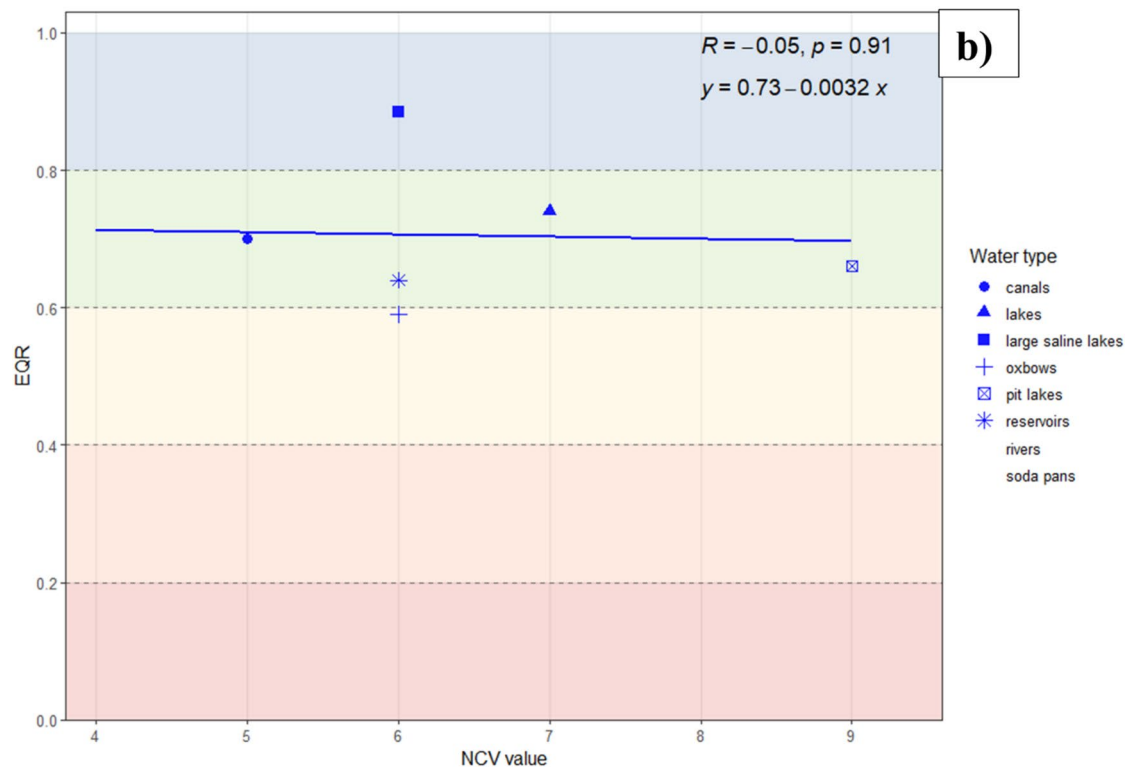
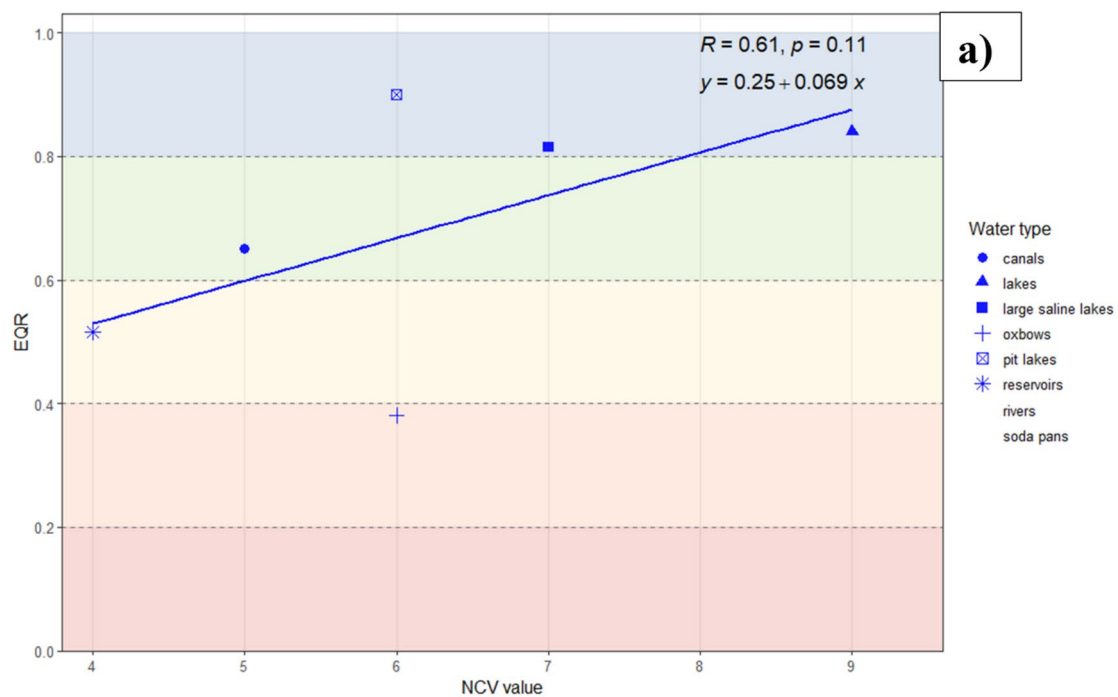


Fig. 2 Relationship between NCV and EQR values at locality level: **a** phytoplankton-based EQR and **b** phyto-benthos-based EQR, with 95% confidence intervals



◀**Fig. 3** Relationship between NCV and EQR values at habitat-type level: **a** phytoplankton-based EQR and **b** phytoenthos-based EQR

NCV showed no discernible relationship with either phytoenthon- or phytoplankton-based EQR values. Although slight positive trends were observed for phytoplankton-based EQR values, the scatter of the data points was considerable, and no clear linear pattern emerged. Accordingly, the coefficients of determination were low for both phytoenthos-based ($R = -0.05$) and phytoplankton-based EQR ($R = 0.61$).

Discussion

Hungarian desmid flora: richness and international comparisons

Desmid diversity data from multiple countries and regions demonstrate substantial variation, with reported species counts ranging between approximately 200 and 400 species depending on geographic scale and sampling intensity. The Czech Republic country-wide survey documented 401 species from over 1400 samples (Šťastný, 2010) which is directly comparable to Hungary's 375 species from 103 localities. From Nigeria, 273 species across multiple locations were reported (Kadiri, 2002). Denmark's Bornholm Island yielded 186 species from four localities (Hansen et al., 2018). Smaller-scale studies reported lower diversity. Austria's Danube flood plain contained 81 species from eight sites (Lenzenweger & Werthl, 2001). Altogether 78 species occurred in a single Hungarian oxbow (Krasznai et al., 2008). Similar number (76 taxa from 18 Czech peat bog sites) has been reported by Štěpánková et al. (2008). Likewise, high desmid diversity has been reported from small ponds and wetlands in the Atlantic region of Western Europe, particularly in the Netherlands (Van Westen & Van Tooren, 2023; 2024) and Belgium (Van Westen & Van Tooren, 2025).

It is well known that desmids exhibit a distinct biogeographical pattern at the continental scale (West, 1909; Krueger, 1933; Krieger, 1937; Coesel, 1996). However, it remains unclear whether the variation observed within continents reflects smaller-scale biogeographical differences, differences in habitat types, or merely differences in sampling effort and

taxonomic expertise. Unfortunately, comprehensive national inventories remain incomplete even in well-studied European regions.

NCV values of the sites and localities

We found that old bog lakes, marshlands, and oxbows show the highest NCV values. However, this was not surprising, as these habitats are hydrologically stable habitats with extended macrophyte coverage, and as to our knowledge showed little changes over time. The Coesel's method was developed for these types of habitats. Although all habitat types contained many localities, only a few could be evaluated for nature conservation purposes. There are three reasons of this: (i) differences in typology, (ii) sampling and (iii) anthropogenic loads.

- (i) Majority of desmids are metaphytic and require a three-dimensional solid physical structure for their survival (Lukács et al., 2024a, 2024b). Littoral macrophyte vegetation provides this habitat for them. Those types such as water currents and astatic soda pans that do not have extended macrophytes in the shoreline usually lack desmids. However, this seems to be contradicted by the fact that some of the large rivers had very high conservation values (ESM 4). The reason is that these rivers themselves are huge collectors, that are connected to canals, oxbows and marshlands (Borics et al., 2014), i.e. to habitats with rich macrovegetation that provide good living conditions for desmid species. In addition, the large rivers are regularly monitored and there is a large amount of data available for them, so there is a high chance that desmid species will be included in some samples. The Danube and Tisza rivers have traditionally been "overrepresented" in research, with a wealth of data collected on them since the twentieth century. Thus, these values characterize the catchment rather than the rivers.

Since most soda pans are highly alkaline and astatic, they are considered hostile environments for desmids. However, some soda pans have relatively stable water balance and extended emergent macrovegetation zones

(Boros et al., 2020), which despite the highly alkaline character of their water allows some desmid species to flourish. The database contained both types, and this has been reflected in the range of NCV values.

In general, we may say that the types considered are fundamentally different from those for which the NCV was originally developed. However, this was partly intentional, as no alternative methods are currently available for assessing the nature conservation value of these habitats.

- (ii) The second reason is the way and frequency of sampling. Collection of desmids would require using plankton net and filtering of water squeezed from macrophytes (Borics et al., 2003). This technique was applied in earlier floristic studies and resulted in a rich desmid flora for several localities even during a simple sampling campaign. However, in the majority of locations, only plankton samples were collected. Likewise, to other metaphytic species, desmids might occur as sporadic elements in the plankton (Görgényi et al., 2019), but their occurrence is highly stochastic and detecting them requires frequent samplings. This was given for the large rivers and also for the large and frequently sampled lakes, i.e. Balaton, Lake Velence and Fertő (Neusiedlersee). This explains their high NCVs.
- (iii) The third reason of the large within type variation of NCV values is the difference in the anthropogenic load. In several standing waters, hydromorphological alterations, high levels of pollution, or fish overstocking (Borics et al., 2012) have reduced or eliminated littoral macrophyte vegetation, which is reflected in the absence of desmids in their reported microflora. However, differences in loads might occur within the types, which explains the high variations of the NCV values.

Relationship between ecological quality (EQR) and nature conservation values (NCVs) at the level of localities

Since the highest (>0.8) Ecological Quality Ratio values should reflect the best quality, characteristic

for undisturbed reference conditions, it is reasonable to suppose that water bodies in this status have also the highest NCV values, and vice versa, low EQRs should coincide with low NCVs. Although our results have not violated the expected monotonic EQR/NCV relationship, they were not convincing. A large range of scatter in NCVs was observed along the EQR gradients calculated both for phytoplankton and phyto-benthos (Fig. 2a, b). Despite the low explained variance and high p-values, two meaningful range in the data distribution could be observed. Namely the data indicated that high NCVs cannot be expected at low (<0.5) phytoplankton EQRs, and small NCVs cannot be expected at high (>0.76) phyto-benthos EQR values (Fig. 2b). The first pattern indicates that the high EQR (>0.5) is a necessary but not sufficient condition for the high NCV. Since our database did not contain localities that could be characterized as poor or bad by the phyto-benthos (EQR:0–0.4), this pattern cannot be identified. The pattern we observed in the phyto-benthos EQR/NCV relation however suggests that at high EQR coincides with high NCV values. The latter is less surprising if we consider that the habitat of benthic diatoms is similar in structure to that of benthic/metaphytic habitat of desmids.

When comparing NCV and EQR values at the level of habitat types we supposed that we would see much lower variance and thus a closer correlation, but this was not the case. The EQRs did not appear to be good predictors of NCVs.

We must note here that the low explanatory power of the model may have partly resulted from unequal sampling intensity among habitat types. The dataset encompasses more than a century of floristic investigations, during which sampling effort has varied considerably among habitat types, with some habitats being surveyed more intensively than others. In addition, differences in the objectives and methodologies of historical floristic investigations, as well as the integration of heterogeneous data sources collected across different periods and spatial scales, may have introduced additional variability into the dataset.

Outlook

The results discussed above shed light on the limitations and challenges of microalgal-based nature conservation value assessment. Developing a useful method requires precisely identifying the

shortcomings of existing approaches and designating the direction of method development. In the following, we will attempt to present these.

Differences between the WFD-compliant methods and those that should be applied for NCV assessment

The expectation for the WFD-compliant methods was that they would provide an evaluable, scalable response to the load on the given biological element (Poikane et al., 2014). Responses to a given pressure can be observed in the structure of communities and in changes in certain quantitative parameters. Quantifying the changes in the relative abundance of the dominant or subdominant taxa within the communities provides a good opportunity to create reliable metrics. The majority of WFD-compliant methods developed by EU member states follow this approach (Birk et al., 2012). During the rehabilitation of highly degraded habitats using these metrics, the direction and rate of changes can be safely tracked. However, the weak relationship between the WFD-compliant metrics and NCVs applied in this study suggests that development of NCV assessment metric requires a different approach. In line with the methodological framework established for the implementation of the WFD, boundaries of NCVs should likewise be linked to clearly defined water body types. Although Coesel (1998) proposed a classification consisting of three groups of water bodies, the complexity and variability of aquatic ecosystems suggest that a more refined typology, incorporating a greater number of distinct water body categories, may be necessary.

The high NCVs of several water bodies are attributable to the unique species they possess. The rarity of a species can be one of the best predictors of its uniqueness. However, this metric should be defined on a clear, quantitative basis, also at the level of the water body types.

The assessment of maturity values and other indicators relevant to species conservation should also be based on mathematical algorithms tailored to the characteristics of individual water body types.

Challenges in sampling and sample processing

Samplings

Differences between approaches used in developing stressor-specific (WFD compliant) and NCV assessment methods should also show differences in the applied sampling strategies. While the ecological status of aquatic ecosystems can be accurately estimated based on the proportion of dominant groups, this is not the case for NCVs. Those techniques that focus on trapping the frequent taxa are not suitable for collecting the rare ones (Carvalho et al., 2012). Detecting rare species in phytoplankton requires a huge investment of time. Therefore, in the case of plankton, only large species can be targeted in samples filtered through plankton nets. This approach was applied by Thunmark (1945) and Järnefelt (1952), and more than seventy years ago. However, the use of benthic or metaphytic algal assemblages or species seems to be more promising. The desmids that are the target group in Coesel's method can be used for this purpose, because these creatures make up the metaphyton, i.e. the assemblage inhabits the lentic environments among the littoral vegetation. Their large size and high relative abundance make them suitable organisms for monitoring purposes. Even in the case of plankton, examining late summer samples can be useful if we want to assess the condition of a water body based on a single sample (Padisák et al., 2003). This may be even more true for benthic and metaphytic communities, which compared to the plankton show a more stable picture over time (Biggs, 1996; Biggs et al., 1998). Several studies demonstrated that examination of water squeezed out of aquatic macrophytes in late summer is particularly useful method and may even be sufficient if we wish to capture the desmid flora of a water body (Borics et al., 2003; Krasznai et al., 2008). Studies on the spatial compositional variability of benthic microalgal assemblages have demonstrated that considerable differences may develop even at small spatial scales (Machová-Černá & Neustupa, 2009; Neustupa et al., 2012). However, potential bias arising from such variability can be reduced by collecting multiple subsamples within each locality. Using this approach, NCV can serve as a robust assessment tool (Neustupa & Němcová, 2025).

Sample processing

The fundamental difference in sample processing is that while WFD-compliant methods seek to determine the exact proportion of species and groups considered dominant based on biomass, determining conservation value requires a floristic survey aimed at confirming the presence or absence of the species in question. Unlike the traditional microscopic sample processing, that aims at determining the relative abundance of 400 species, relative frequencies are not important in floristic surveys, only the detection of presence, but this may require examining a significantly larger number of individuals. Unfortunately, there are no good practices in the literature on how this can be standardized. We note here that both sampling and sample processing difficulties can be reduced by the application of molecular species identification in the future, as it has already been successfully applied in the case of benthic diatoms (Rivera et al., 2018).

Size problems

As the WFD only requires member states to examine and evaluate standing waters larger than 50 hectares, smaller ponds and wetlands are not covered by the monitoring activities. However, areas of high conservation value may be much smaller than this. While the impact of the most important stressors, i.e. organic loads and the nutrients on the aquatic ecosystems does not depend on the size of the lakes, the species richness is a size-dependent variable (Borics et al. 2016; Várbiro et al., 2017). Therefore, before using the species richness to quantify NCV of a habitat, a typology should be created where the size should be an important type descriptor variable. Failure to account for water body size may lead to systematically lower NCV values for small water bodies, thereby causing them to be classified as habitats of lower ecological value. However, at regional level, the aggregated NCVs of small waters can be very high. Studying the SLOSS debate (single large or several small) for microalgae, Bolgovics et al. (2019) proved that aggregated species diversity of small waterbodies was considerably higher than that of a large water body with the same surface area.

Involvement of other microalgal groups

If we want to develop a microalgal-based method that is suitable for determining the conservation value of a habitat, the selected algae group must have the following characteristics. They must be sampled and identified easily, and they also must possess a type-specific occurrence pattern. The desmids indisputably meet these criteria, but for the other groups it is not necessarily true. There are many silica-scaled chrysophytes that are characteristic for some water types (Péterfi et al., 1998; Padisák et al., 2000) but the identification of species is not a simple task. Similarly, this is true for the diatoms and dinoflagellates. The typical pond dwellers like euglenophytes (Pringsheim, 1956) seem like a good choice. They occasionally have large diversity even in a small water body and compared to species of the above-mentioned group, their identification is much easier. Of course, the question remains: How can their indicator values be expressed objectively and quantitatively?

However, differences in NCV values among the habitat types indicate that the habitats should have a type-specific range of these values. This practically means that similarly as it has been done when Water Framework Directive compliant assessment methods were developed, type-specific reference assemblages should be set for each water body type.

Conclusions

During the restoration of nature conservation areas, biological indicators are needed to monitor the success of interventions. The currently used WFD-compliant metrics that aim to quantify the impact of specific stressors are of limited or no use for estimating the conservation value of a water body. Despite its simplicity, Coesel's framework appears to be a promising evaluation system. The shortcomings associated with regional specificity and subjectivity could likely be addressed through methodological refinement, leading to the development of a more robust and reliable assessment metric.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval No human participants and animals were involved in the research.

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